Fuzzy Logic + Uncertainty

Short-Notes Questions & Model Answers

Past Paper Questions & Solutions (2020–2024)

CSE 713 Artificial Intelligence

Study Notes — June 2026

Contents

Exam Pattern Summary

Key Exam Facts

- **Yield: only 2 of the 5 years (2020–2024) ask it** — 2024 and 2022. Lowest-yield topic in the syllabus.
- **Always appears the same way:** a 2–3 mark "write short notes" sub-part, bundled *inside* the Neural Networks question (never its own question).
- **Always paired with Uncertainty** — the two short notes are asked together every single time they appear.
- **2021 and 2020 do NOT ask it at all** — the course's question map previously listed them as "combined with NN", but the raw papers contain no Fuzzy Logic mention in those years (corrected 2026-06-07).
- **Strategy:** 10–15 minutes max. Memorise the one-paragraph definition + 4-step process below; do not deep-dive into COG arithmetic (never asked as a numerical in any year so far).

Map Correction Note

The original `_TopicQuestionMap.md` listed 2024 under Section A Q4 and claimed 2021/2020 had "combined with NN" Fuzzy Logic sub-parts. Both raw papers were re-checked page by page: **2021 and 2020 contain no Fuzzy Logic question whatsoever** (their NN-adjacent Section B questions cover only Bayesian Networks / Uncertainty / Meningitis, not Fuzzy Logic). The 2024 question is in **Section B, Q8c** (paired with the NN backprop numerical in Q8b), not Section A. The map has been corrected.

2024 — Q8c (3 marks)

2024 Section B, Q8c

Write short notes on the following:

i) Fuzzy logic

ii) Uncertainty

(Asked as the closing sub-part of Q8, whose main body (Q8a, Q8b) covered associative memory and a sigmoid-network backpropagation numerical — see `NeuralNet_Solutions.tex` / `BayesNet_Solutions.tex` for those.)

Solution — (i) Fuzzy Logic

Definition (exam-ready, one sentence):
Fuzzy Logic (FL), conceived by Prof. Lotfi Zadeh in the mid-1960s, is a problem-solving methodology that simulates the process of normal human reasoning under *uncertainty* by allowing degrees of truth between 0 and 1, instead of the strict true/false of Boolean logic.

Why it exists — Boolean vs. Fuzzy sets:

- **Conventional (crisp) set:** membership is all-or-nothing — $f_S(x) \in \{0, 1\}$. The transition between "not a member" and "member" is *instantaneous* (e.g., 79°F is either "warm" or not).
- **Fuzzy set:** membership is graded — a *membership function* $m_A(x)$ maps each element to a value in $[0, 1]$, so the transition is *gradual* and an object can partially belong to multiple sets at once (e.g., 80°F can be 0.6 "warm" AND 0.3 "hot" simultaneously).

$$m_A(x) = 1 \Rightarrow x \text{ totally in } A, \quad 0 < m_A(x) < 1 \Rightarrow x \text{ partially in } A, \quad m_A(x) = 0 \Rightarrow x \text{ not in } A$$

Five basic concepts of a fuzzy set: Label, Degree of membership, Membership function, Scope (domain), Universe of discourse.

The Fuzzy Decision Process — 3 steps (the core of any short note):

1. **Fuzzification:** crisp input → fuzzy input. Assign fuzzy linguistic labels (e.g., cold/cool/normal/warm/hot) to the input via membership functions (commonly trapezoidal or triangular). E.g., crisp input $T = 66$°F gives two fuzzy inputs: $T_{cool} = 0.2$, $T_{normal} = 0.6$.
2. **Fuzzy reasoning (Rule Evaluation), via min–max inference:** apply IF–THEN rules of the form `IF <antecedent> AND <antecedent> THEN <consequent>`. Rule strength = min of antecedent truth values (fuzzy AND = intersection); when several rules share a consequent, the fuzzy output = max of their strengths (fuzzy OR = union). Fuzzy NOT (complement): $m_{\bar{A}}(x) = 1 - m_A(x)$.
3. **Defuzzification:** fuzzy output → crisp output. The most common technique

is the **Centre of Gravity (COG) / centroid method**:

$$COG = \frac{\int_a^b m(x)\, x\, dx}{\int_a^b m(x)\, dx} \approx \frac{\sum_{x=a}^{b} x \cdot m(x)}{\sum_{x=a}^{b} m(x)}$$

(For singleton outputs, $COG = \dfrac{\sum_i FO_i \times SPos_i}{\sum_i FO_i}$, which needs far fewer computations.)

Advantages (pick 3–4 if asked): provides flexibility; models uncertainty; shortens system development time; increases maintainability; uses cheaper hardware; handles control/decision problems that are hard to define mathematically.
Applications (pick 2–3 examples): antilock braking systems, washing-machine control, air conditioners, train motion control, photocopier exposure control (Matsushita), strategic planning, stock selection.
One-line closer if more space is needed: Fuzzy systems are one of the three pillars of *Soft Computing* (with Neural Networks and Genetic Algorithms); combining FL with NN gives *neuro-fuzzy systems*, which let a network learn/tune the membership functions and rules that a pure fuzzy system cannot acquire automatically.

Solution — (ii) Uncertainty

Definition: Uncertainty arises whenever an agent must reason or act on incomplete, imprecise, noisy, or conflicting information — situations where classical sound inference (deduction) cannot guarantee a correct conclusion.
Canonical exam example — the doorbell:

- Prop 1: $AtDoor(x) \rightarrow Doorbell$
- Prop 2: $Doorbell \rightarrow Wake(Karim)$
- The doorbell rings at midnight — was someone at the door? Did Karim wake up? **Neither can be concluded with certainty**: the doorbell could have a fault (rang with no one there), and Karim might sleep through it. Both forward (deductive) and backward (abductive) chaining fail to give a sound guarantee.

Why deduction/abduction are not "sound" here: they require complete and certain premises; in the real world, premises are themselves uncertain, so the conclusion inherits that uncertainty (affirming the consequent is not valid).
Sources of uncertainty (list 3–4): weak implications/rules of thumb, imprecise language ("often", "rarely"), incomplete knowledge of the domain, conflicting evidence from multiple sources, and propagation/compounding of uncertainty through chains of inference.
How AI handles it: probability theory and **Bayes' Theorem** (see `BayesNet_Solutions.tex` for the full doorbell write-up, the Extended Bayes forms, and Bayesian Network worked examples — that file is the authoritative source for any numerical Uncertainty/Bayes question).

2022 — B-1d (2 marks)

2022 Section B, B-1d

Write short notes on the following:

i) Fuzzy logic

ii) Uncertainty

(Closing sub-part of B-1, whose main body (B-1a, B-1b, B-1c) asked: what is an ANN & how it compares to biological networks; what is learning / inductive learning; and supervised vs. unsupervised vs. reinforcement learning — see `NeuralNet_Solutions.tex` *for those.)*

Solution

Identical question to 2024 Q8c, worth slightly fewer marks (2 vs. 3) — use the exact same two model answers above, trimmed to fit:

- **Fuzzy logic (condensed):** A methodology (Zadeh, 1960s) that lets truth be a matter of degree in $[0, 1]$ via membership functions, instead of strict true/false. Process = Fuzzification → Fuzzy reasoning (min–max inference on IF–THEN rules) → Defuzzification (Centre of Gravity). Used for control/decision problems that resist precise mathematical modelling (e.g., washing machines, ABS brakes).
- **Uncertainty (condensed):** Arises when information is incomplete, imprecise, or conflicting, so deduction/abduction cannot guarantee a correct conclusion (doorbell-at-midnight example). AI manages it quantitatively via probability theory and Bayes' Theorem.

Exam tip: with only 2 marks on offer, one crisp sentence per term plus one example each is sufficient — do not write the full process breakdown unless marks ≥ 3.

Quick Reference Summary

Year-by-Year Appearance (2020–2024)

Year	Location	Ask
2024	Section B, Q8c (3 marks)	Short notes: Fuzzy logic + Uncertainty (inside NN question)
2023	—	Not asked
2022	Section B, B-1d (2 marks)	Short notes: Fuzzy logic + Uncertainty (inside NN question)
2021	—	Not asked (no NN/Fuzzy question that year)
2020	—	Not asked (no NN/Fuzzy question that year)

The One Paragraph to Memorise Cold

"Fuzzy Logic (Zadeh, 1960s) is a problem-solving methodology that simulates human reasoning under uncertainty using the mathematical theory of fuzzy sets, where membership grades range continuously over $[0, 1]$ rather than being strictly 0 or 1 as in Boolean logic. A fuzzy decision process has three steps: **Fuzzification** (crisp input → fuzzy input via membership functions), **Fuzzy reasoning** (apply IF–THEN rules using min–max inference: AND = minimum, OR = maximum), and **Defuzzification** (fuzzy output → crisp output, typically via the Centre-of-Gravity / centroid method). It is used in control and decision-making systems that are difficult to model mathematically, e.g., ABS brakes, washing machines, photocopier exposure control."

Five Things That Could Be the +1 Mark

1. Boolean logic = instantaneous transition; Fuzzy logic = gradual transition (partial membership in multiple sets).
2. Fuzzy AND → minimum (intersection); Fuzzy OR → maximum (union); Fuzzy NOT → $1 - m_A(x)$ (complement).
3. Trapezoidal and triangular shapes are the most common membership functions; singletons simplify defuzzification.
4. COG formula: $COG = \dfrac{\sum x \cdot m(x)}{\sum m(x)}$ — worked exam value from slides: $COG = 34.5$.
5. Fuzzy Logic + Neural Networks + Genetic Algorithms = **Soft Computing**; combining FL and NN gives **neuro-fuzzy systems** (cooperative: NN tunes FL parameters; hybrid: FL rules implemented as a fuzzy-neuron network).

Common Mistakes

- Don't confuse *fuzzification* (crisp→fuzzy, step 1) with *defuzzification* (fuzzy→crisp, step 3) — a frequent slip under time pressure.
- Min–max is for *rule evaluation / reasoning* (step 2), NOT for defuzzification (which uses Centre of Gravity).
- Don't write a generic "what is uncertainty" answer without the doorbell example or a source-of-uncertainty list — examiners reward the named example.
- This topic is low-yield (2/5 years) — don't over-invest revision time relative

to Bayes/BN, NN, or FOL, all of which appear every year.